Michelle Huang

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Summary

Fifth-year doctoral candidate working on mastering the foundational skills and knowledge to ask thoughtful questions and design experiments. My research interests in vector control, neglected tropical diseases, and health in humanitarian emergencies are heavily influenced by my public health background and desire to serve under-resourced communities.

Education

University of Notre Dame **Aug 2020-present**

Fifth-Year Doctoral Candidate, Integrated Biomedical Sciences, Immunology and infectious diseases

- Coursework in: molecular biology, biocomputing, immunology, epidemiology, population genetics, global health
- 4.00/4.00 Grade Point Average (GPA)

Johns Hopkins Bloomberg School of Public Health **Nov 2017-Dec 2019**

Master of Public Health, with Certificate in Tropical Medicine

- Coursework in: epidemiology, biostatistics, clinical trials, neglected tropical diseases, qualitative research, program planning, immunology, analysis of biological sequences, American Indian health disparities
- Capstone: Seroprevalence of *Trypanosoma cruzi* in American Dogs—A Sentinel for Human Infection
- 3.95/4.00 GPA

Saddleback College **Apr 2015-Nov 2015**

Paramedic Certificate

- Finished 3rd in paramedic class #75
- 4.0/4.0 GPA

Miramar College **Jan 2012-March 2014**

Certificate in Fire Protection Technology, Highest Honors

- 4.0/4.0 GPA

University of California, San Diego **Sept 2010-June 2012**

BA in Visual Arts, Department Honors with High Distinction

- Minor in psychology
- Nomination for- and completion of the Studio Arts Honors Program
- 3.63/4.0 GPA

Saddleback College/Irvine Valley College **July 2009-June 2010**

AA Social and Behavior Sciences, cum laude; IGETC certification

- Saddleback: 4.0/4.0 GPA
- Irvine Valley College: 3.6/4.0 GPA

Work and Volunteer Experience

University of Notre Dame **Aug 2020-present**

Leishmaniasis and Chagas Laboratory, Research Student (40+ hrs/wk)

- Sand fly genomics and metagenomics
  - Updated reference genome assemblies of two vectors of leishmaniasis, *Lutzomyia longipalpis* and *Phlebotomus papatasi*; prepared and published scientific report on sequences
  - Screened and tested a panel of genetic age-markers to reliably identify the age of a sand fly
  - Used Whole Genome Sequencing (WGS) data for analysis and interpretation of *L. longipalpis* species complex population genetics
  - Conducted WGS and 16s rRNA sequencing to analyze microbial composition of experimentally infected sand flies and identify core taxa shared in common with wild populations to illuminate transmission dynamics
  - Gave talks and presented findings at various scientific conferences and fora
  - Gained experience in research; data collection, analysis and interpretation; data and information synthesis from various databases
- Triatomine and *Trypanosoma cruzi* diagnostics
  - Conducted restriction-site associated DNA sequencing to identify genetic markers to differentiate between different species of triatomine bugs, and between *T. cruzi* and its non-pathogenic sister species, *T. rangeli*
  - Gained experience in research; data analysis and interpretation; and collaboration and partnering
- Mosquito surveillance
  - Led, trained, and mentored 17 undergraduate and high school students in a pilot collaborative public health project between the College of Sciences and the St. Joseph County Department of Health for four years
  - Set traps, collected and identified 184,571 mosquitoes, sent disease-carrying species for viral testing, and identified two West Nile Virus positive mosquito pools to inform communicable disease surveillance, planning, and intervention strategies
  - Conducted community engagement and education events on mosquito-borne diseases

- Developed protocols, contributed data, and validated use of novel AI technology, Vectech, for mosquito sorting
- Built predictive models of mosquito abundance based on weather variables
- Gained experience in project management; public health knowledge; collaboration and partnering; needs assessments; and data collection, analysis, and interpretation

Memorial Hospital **Jan 2022-Feb 2025**
Emergency Care Center, Paramedic (24 hrs/mo)
- Perform patient care, and task and triage in a level II trauma center
- Observe proper protocols for prevention, control, and treatment of communicable diseases
- Compliance with public health policies, public health laws, and HIPAA

Elkhart General Hospital **Nov 2020-Nov 2022**
Emergency Department, Paramedic (24 hrs/mo)
- Performed patient care, and task and triage in a level III trauma center

Team Rubicon **2018-present**
WASH Team, Team Member (variable, 0-30 hrs/wk) Feb 2023-present
- Coordinate WASH (water, sanitation, hygiene) pack-out for impending and future deployments (Ukraine, Malawi)
- Participate in WASH trainings and needs discussions
- Assist with World Health Organization (WHO) Emergency Medical Team (EMT) Type I Mobile reclassification process

Program Design and Learning, Preparedness Research Specialist (20hrs/wk) Jan 30-Mar 31, 2023
- Researched and outlined a digital information resource for all-hazard preparedness, response, and recovery for remote, and low-resource communities

International Team Selection Course, Cadre (continuous, 40+ hrs/wk) Oct 16-Oct 23, 2022
- Coordinated logistics, functioned as role-player and team-member, completed analysis on candidate performance

Calling Gloria, Volunteer (40 hrs/wk) Aug 17-Aug 23, 2022
- Functioned as Situation Unit Leader and general responder

International Team Selection Course, Participant (continuous, 40+ hrs/wk) Dec 7-Dec 11, 2021
- Piloted selection course for response on a WHO EMT Type I Mobile unit
- Functioned as Paramedic and Medical Unit Lead
- Rostered for immediate deployment in response to humanitarian emergencies

Resolute Sentinel-21 Guatemala deployment, Paramedic (continuous, 40+ hrs/wk) June 18-July 4, 2021
- Performed triage, patient care and education duties and staffed the pharmacy
- Completed needs assessments, staff education, and COVID-19 vaccinations

Johns Hopkins Bloomberg School of Public Health **Oct 2019-Dec 2019**
Chagas Laboratory, Research Student (20 hrs/wk)
- Performed indirect immunofluorescent assays for *T. cruzi* diagnostics
- Assisted with grant and proposal writing
- Facilitated collaboration across institutions and tribal government for design of an original study

Universidad Peruana Cayetano Heredia/Johns Hopkins **Feb 2019-May 2019**
Comparative Bacterial Genomics Laboratory, Research Student (40 hrs/wk)
- Performed sample collection, bacteria culturing, biochemical and disk diffusion tests, DNA extraction and quantification
- Contributed to study design, protocol writing, manuscript drafting for multiple projects

Pueblo Community College **Jan 2018-Feb 2019**
EMT-B Instructor(20 hrs/wk)
- Created lesson plans, lectured, taught skills, proctored exams, and set up multidisciplinary, multi-agency scenarios

Mercy Regional Medical Center **Jan 2017-Feb 2019**
Emergency Room Paramedic (20 hrs/wk)
- Performed patient care, and task and triage in a level III trauma center

San Juan Regional Medical Center **Apr 2016-Oct 2016**
Lead Paramedic (40+ hrs/wk)
- Performed patient care and transport for 911 emergencies; conducted scene and resource management
- Mentored and managed an EMT-B or EMT-I partner

Mercy Regional Medical Center **Sept 2014-Apr 2015**
Patient Care Technician (20 hrs/wk)
- Performed patient care, task and triage

Nursing Unit Secretary (20 hrs/wk)
- Coordinated and managed daily activities and resources of emergency department

Miramar College **Nov 2012-Apr 2015**
EMT-B Instructor (20 hrs/wk)
- Taught skills, advised during open labs, proctored skills exams

San Diego County Fire Authority **Sept 2011-May 2013**
Volunteer/Reserve Firefighter (48 hrs/mo)

- Responded to all-hazards, including fire, medical, public assists, and other miscellaneous public service calls
- Conducted training and community education

UCSD Police Department **Feb 2011-June 2012**
Community Service Officer (20 hrs/wk)

- Performed campus escorts, public assists, and functioned as medical at special events

Posters and Publications

Murray M, Salvatierra G, Dávila-Barclay A et al. (2021). Market Chickens as a Source of Antibiotic-Resistant Escherichia coli in a Peri-Urban Community in Lima, Peru. *Frontiers in microbiology*, *12*, 327.
Huang M, Kingan S, Shoue D, et al. (2024). Improved high quality sand fly assemblies enabled by ultra low input long read sequencing. *Sci Data*, 11(1):918. Published 2024 Aug 24. doi:10.1038/s41597-024-03628-y
D'Antuono, C., Anderson, K., Afuso, J et al. (2024). Mosquito abundance and species surveillance in St. Joseph County, Indiana, 1976-1997. *ARPHA Preprints*, *5*, e122515.
Huang M, Dimpel J, Zeng Z et al. Identification of age-markers for the sand fly vector of *Leishmania major*, *Phlebotomus papatasi*. Poster presented at: the annual Midwest Neglected Infectious Diseases meeting; Aug 12, 2022; Notre Dame, IN
Huang M, Bruzzese D, Shoue D et al. Metagenomics of *Phlebotomus papatasi* and *Lutzomyia longipalpis*. Poster presented at: the annual American Society of Tropical Medicine and Hygiene meeting; Oct 30, 2022; Seattle, WA.
Huang M, Bruzzese D, Shoue D et al. Population genetics and metagenomics of the sand fly vector of Leishmania using updated genome assemblies of Phlebotomus papatasi and Lutzomyia longipalpis. Poster presented at: the annual American Society of Tropical Medicine and Hygiene meeting; Oct 18, 2023; Chicago IL.

References

Mary Ann McDowell, Professor and Associate Dean for Professional Development; University of Notre Dame (mmcdowe1@nd.edu)
Jennifer Robichaud, Teaching Professor; University of Notre Dame (jrobicha@nd.edu)
Cristian Koepfli, Assistant Professor; University of Notre Dame (ckoepfli@nd.edu)
Robert Gilman, Professor; Johns Hopkins University (rgilman1@jhmi.edu)